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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

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In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

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may indicates permission to do something

need not indicates permission not to do something

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can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document specifies the usage of the Unified Data Repository services for Ambient IoT Data. It provides the resource definition and data model for Ambient IoT data used over the Nudr Service Based Interface. The protocol definition for Nudr Service Based Interface which is specific to Ambient IoT data is also specified in the present document.

The 5G System stage 2 architecture and procedures are specified in TS 23.501 [2], TS 23.502 [3] and TS 23.369 [14].

The stage 3 protocol definition for Nudr Service Based Interface which is common for subscription data, policy data, structure data for exposure, application data and Ambient IoT data is specified in TS 29.504 [15].

The Technical Realization of the Service Based Architecture and the Principles and Guidelines for Services Definition are specified in TS 29.500 [4] and TS 29.501 [5].

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
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[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 23.501: "System Architecture for the 5G System; Stage 2".

[3] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".

[4] 3GPP TS 29.500: "5G System; Technical Realization of Service Based Architecture; Stage 3".

[5] 3GPP TS 29.501: "5G System; Principles and Guidelines for Services Definition; Stage 3".

[6] OpenAPI: "OpenAPI Specification Version 3.0.0", https://spec.openapis.org/oas/v3.0.0.

[7] 3GPP TR 21.900: "Technical Specification Group working methods".

[8] 3GPP TS 33.501: "Security architecture and procedures for 5G system".

[9] IETF RFC 6749: "The OAuth 2.0 Authorization Framework".

[10] 3GPP TS 29.510: "5G System; Network Function Repository Services; Stage 3".

[11] IETF RFC 9113: "HTTP/2".

[12] IETF RFC 8259: "The JavaScript Object Notation (JSON) Data Interchange Format".

[13] IETF RFC 9457: "Problem Details for HTTP APIs".

[14] 3GPP TS 23.369: "Architecture support for Ambient power-enabled Internet of Things;Stage 2".

[15] 3GPP TS 29.504: "5G System; Unified Data Repository Services; Stage 3".

[16] IETF RFC 9110: "http Semantics".

[17] IETF RFC 9111: "HTTP Caching".

[18] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces Stage 3".

[19] 3GPP TS 29.369: "Ambient IoT Data Management Services".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [1] and TS 23.369 [14] and the following apply. A term defined in the present document or in TS 23.369 [14] takes precedence over the definition of the same term, if any, in TR 21.905 [1]. A term defined in the present document takes precedence over the definition of the same term, if any, in TS 23.369 [14].

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and TS 23.369 [14] and the following apply. An abbreviation defined in the present document or in TS 23.369 [14] takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1]. An abbreviation defined in the present document takes precedence over the definition of the same term, if any, in TS 23.369 [14].

4 Overview

The resources, data model and usage defined in the present document are used by the 5GC NFs (e.g., ADM) to access the Ambient IoT data via the Unified Data Repository services specified in TS 29.504 [15].

5 Usage of Nudr_DataRepository Service API for Ambient IoT data

5.1 Introduction

The following clauses specify the usage of Nudr_DataRepository service for Ambient IoT data. The principles specified in TS 29.504 [15] are followed unless explicitly specified otherwise in the following clauses.

5.2 Resources

5.2.1 Overview

This clause specifies the resources for the usage of the Unified Data Repository service for Ambient IoT data.

5.2.2 Resource Structure

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 5.2.2-1 depicts the resource URIs structure for the Nudr_DataRepository API for Ambient IoT data.

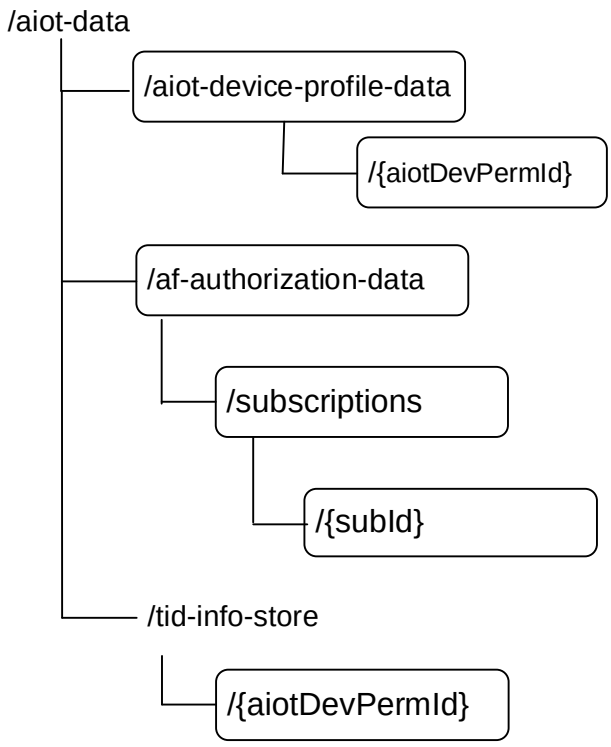


Figure 5.2.2-1: Resource URI structure of the Nudr_DataRepository API for Ambient IoT data

Table 5.2.2-1 provides an overview of the resources and applicable HTTP methods.

Table 5.2.2-1: Resources and methods overview

Resource name	Resource URI	HTTP method or custom operation	Description
AiotDeviceProfileData	/aiot-data/aiot-device-profile-data/{aiotDevPermId}	GET	Retrieve the AIoT device profile data for an AIoT device permanent identifier.
		PATCH	Modify the AIoT device profile data for an AIoT device permanent identifier.
BundledAiotDeviceProfileData	/aiot-data/aiot-device-profile-data	GET	Retrieve the AIoT device profile data for bundled AIoT devices.
		PATCH	Modify the AIoT device profile data for bundled AIoT devices.
AfAuthorizationData	/aiot-data/af-authorization-data	GET	Retrieve AF Authorization Data
AfAuthorizationDataSubscriptions	/aiot-data/af-authorization-data/subscriptions	POST	Create a subscription of AF Authorization Data change
Individual AfAuthorizationDataSubscription	/aiot-data/af-authorization-data/subscriptions/{subId}	PATCH	Modify a subscription of AF Authorization Data change
		DELETE	Delete a subscription of AF Authorization Data change
IndividualTidInformation	/tid-info-store/{aiotDevPermId}	GET	Retrieve the stored T-ID information for an AIoT device.
		PATCH	Update the stored T-ID information for an AIoT device.

5.2.3 Resource: AiotDeviceProfileData

5.2.3.1 Description

The resource represents all AIoT device profile data attributes in the UDR for a given "aiotDevPermId".

5.2.3.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/aiot-data/aiot-device-profile-data/{aiotDevPermId}**

This resource shall support the resource URI variables defined in table 5.2.3.2-1.

Table 5.2.3.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See TS 29.504 [15] clause 6.1.1
aiotDevPermId	AiotDevPermId	Represents the AIoT device permanent id. Data type AiotDevPermId is defined in TS 29.571 [18].

5.2.3.3 Resource Standard Methods

5.2.3.3.1 GET

This method shall support the URI query parameters specified in table 5.2.3.3.1-1.

Table 5.2.3.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.3.3.1-2 and the response data structures and response codes specified in table 5.2.3.3.1-3.

Table 5.2.3.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.3.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AiotDevProfileData	M	1	200 OK	Upon success, a response body containing AIoT Device Profile Data shall be returned.
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of TS 29.500 [4] also apply.				

5.2.3.3.2 PATCH

This method shall support the URI query parameters specified in table 5.2.3.3.2-1.

Table 5.2.3.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
supported-features	SupportedFeatures	O	0..1	The features supported by the NF service consumer.

This method shall support the request data structures specified in table 5.2.3.3.2-2 and the response data structures and response codes specified in table 5.2.3.3.2-3.

Table 5.2.3.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
array(PatchItem)	M	1..N	Contains the delta data to the provisioned parameter data

Table 5.2.3.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon successful modification there is no body in the response message. (NOTE 2)
PatchResult	M	1	200 OK	Upon success, the execution report is returned. (NOTE 2)
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE 1: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of TS 29.500 [4] also apply. NOTE 2: If all the modification instructions in the PATCH request have been implemented, the UDR shall respond with 204 No Content response; if some of the modification instructions in the PATCH request have been discarded, and the NF service consumer has included in the supported-feature query parameter the "PatchReport" feature number, the UDR shall respond with PatchResult.				

5.2.4 Resource: AfAuthorizationData

5.2.4.1 Description

This resource represents the AF Authorization data.

This resource is modelled as the Document resource archetype (see clause C.1 of TS 29.501 [5]).

5.2.4.2 Resource Definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/aiot-data/af-authorization-data**

This resource shall support the resource URI variables defined in table 5.2.4.2-1.

Table 5.2.4.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See TS 29.504 [15] clause 6.1.1

5.2.4.3 Resource Standard Methods

5.2.4.3.1 GET

This method shall support the URI query parameters specified in table 5.2.4.3.1-1.

Table 5.2.4.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description	Applicability
af-id	AfId	O	0..1	When present, this parameter indicates the AF for whom the Authorization data is to be retrieved. When this IE is absent, it shall indicate that the Authorization Data for all the AFs are to be retrieved.	

This method shall support the request data structures specified in table 5.2.4.3.1-2 and the response data structures and response codes specified in table 5.2.4.3.1-3.

Table 5.2.4.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.4.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AfAuthorizationData	M	1	200 OK	Upon success, a response body containing the AfAuthorizationData shall be returned.
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE: The mandatory HTTP error status code for the GET method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.				

5.2.4.4 Resource Custom Operations

None.

5.2.5 Resource: BundledAiotDeviceProfileData

5.2.5.1 Description

The resource represents a collection resource of different AiotDeviceProfileData with the resource URI "aiotDevPermId".

5.2.5.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/aiot-data/aiot-device-profile-data**

This resource shall support the resource URI variables defined in table 5.2.5.2-1.

Table 5.2.5.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See TS 29.504 [15] clause 6.1.1

5.2.5.3 Resource Standard Methods

5.2.5.3.1 GET

This method shall support the URI query parameters specified in table 5.2.5.3.1-1.

Table 5.2.5.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
requester-aiot-devices-id	array(AiotDevPermId)	M	1..N	This IE shall contain a list of Requested AIoT device permanent identifier.

This method shall support the request data structures specified in table 5.2.5.3.1-2 and the response data structures and response codes specified in table 5.2.5.3.1-3.

Table 5.2.5.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.5.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AiotDevProfileDataList	M	1	200 OK	Upon success, a response body containing AIoT Device Profile Data for bundled AIoT Devices shall be returned.
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of TS 29.500 [4] also apply.				

5.2.5.3.2 PATCH

This method shall support the URI query parameters specified in table 5.2.5.3.2-1.

Table 5.2.5.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
supported-features	SupportedFeatures	O	0..1	The features supported by the NF service consumer.

This method shall support the request data structures specified in table 5.2.3.3.2-2 and the response data structures and response codes specified in table 5.2.3.3.2-3.

Table 5.2.5.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
array(PatchItem)	M	1..N	It contains the list of changes to be made to the AIoT Device Profile Data for bundled AIoT Devices, according to the JSON PATCH format specified in IETF RFC 6902 [13].

Table 5.2.5.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon successful modification there is no body in the response message. (NOTE 2)
PatchResult	M	1	200 OK	Upon success, the execution report is returned. (NOTE 2)
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE 1: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of TS 29.500 [4] also apply. NOTE 2: If all the modification instructions in the PATCH request have been implemented, the UDR shall respond with 204 No Content response; if some of the modification instructions in the PATCH request have been discarded, and the NF service consumer has included in the supported-feature query parameter the "PatchReport" feature number, the UDR shall respond with PatchResult.				

5.2.6 Resource: AfAuthorizationDataSubscriptions

5.2.6.1 Description

This resource is used to represent the collection of Subscriptions for AF Authorization Data Changes.

5.2.6.2 Resource Definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/aiot-data/af-authorization-data/subscriptions**

This resource shall support the resource URI variables defined in table 5.2.6.2-1.

Table 5.2.6.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 6.1.1

5.2.6.3 Resource Standard Methods

5.2.6.3.1 POST

This method shall support the URI query parameters specified in table 5.2.6.3.1-1.

Table 5.2.6.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description	Applicability

This method shall support the request data structures specified in table 5.2.6.3.1-2 and the response data structures and response codes specified in table 5.2.6.3.1-3.

Table 5.2.6.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
AfAuthorizationDataSubscription	M	1	Contain the information of the AF Authorization Data Subscription to be created.

Table 5.2.6.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AfAuthorizationDataSubscription	M	1	201 Created	Upon success, a response body containing the AF Authorization Data Subscription shall be returned.
NOTE: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.				

Table 5.2.6.3.1-4: Headers supported by the 201 response code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created subscription, according to the structure: {apiRoot}/nudr-dr/<apiVersion>/aiot-data/af-authorization-data/subscriptions/{subId}

5.2.7 Resource: Individual AfAuthorizationDataSubscriptions

5.2.7.1 Description

This resource is used to represent an Individual Subscription of AF Authorization Data Changes.

5.2.7.2 Resource Definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/aiot-data/af-authorization-data/subscriptions/{subId}**

This resource shall support the resource URI variables defined in table 5.2.7.2-1.

Table 5.2.7.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 6.1.1
{subId}	string	Subscription Id

5.2.7.3 Resource Standard Methods

5.2.7.3.1 PATCH

This method shall support the URI query parameters specified in table 5.2.7.3.1-1.

Table 5.2.7.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description	Applicability

This method shall support the request data structures specified in table 5.2.7.3.1-2 and the response data structures and response codes specified in table 5.2.7.3.1-3.

Table 5.2.7.3.1-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
array(PatchItem)	M	1..N	Contain the patch instructions to be applied to the subscription.

Table 5.2.7.3.1-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
PatchResult	M	1	200 OK	Upon partial success, a response body containing the patch result shall be returned.
n/a			204 No Content	Upon success, 204 response code shall be returned.
NOTE: The mandatory HTTP error status code for the PATCH method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.				

5.2.7.3.1 DELETE

This method shall support the URI query parameters specified in table 5.2.7.3.1-1.

Table 5.2.7.3.1-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description	Applicability

This method shall support the request data structures specified in table 5.2.7.3.1-2 and the response data structures and response codes specified in table 5.2.7.3.1-3.

Table 5.2.7.3.1-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.7.3.1-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a.			204 No Content	Upon success, 204 response code shall be returnd.
NOTE: The mandatory HTTP error status code for the DELETE method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.				

5.2.8 Resource: IndividualTidInformation

5.2.8.1 Description

The resource represents an individual stored T-ID Information in the UDR for a given "aiotDevPermId".

5.2.8.2 Resource definition

Resource URI: **{apiRoot}/nudr-dr/<apiVersion>/aiot-data/tid-info-store/{aiotDevPermId}**

This resource shall support the resource URI variables defined in table 5.2.8.2-1.

Table 5.2.8.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See TS 29.504 [15] clause 6.1.1
aiotDevPermId	AiotDevPermId	Represents the AIoT device permanent id. Data type AiotDevPermId is defined in TS 29.571 [18].

5.2.8.3 Resource Standard Methods

5.2.8.3.1 GET

This method shall support the URI query parameters specified in table 5.2.8.3.1-1.

Table 5.2.8.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.2.8.3.1-2 and the response data structures and response codes specified in table 5.2.8.3.1-3.

Table 5.2.8.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.2.8.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
TidInformation	M	1	200 OK	Upon success, a response body containing the stored T-ID information shall be returned.
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of TS 29.500 [4] also apply.				

5.2.8.3.2 PATCH

This method shall support the URI query parameters specified in table 5.2.8.3.2-1.

Table 5.2.8.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
supported-features	SupportedFeatures	O	0..1	The features supported by the NF service consumer.

This method shall support the request data structures specified in table 5.2.8.3.2-2 and the response data structures and response codes specified in table 5.2.8.3.2-3.

Table 5.2.8.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
array(PatchItem)	M	1..N	Contains the delta data to the provisioned parameter data

Table 5.2.8.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon successful modification there is no body in the response message. (NOTE 2)
PatchResult	M	1	200 OK	Upon success, the execution report is returned. (NOTE 2)
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE 1: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of TS 29.500 [4] also apply. NOTE 2: If all the modification instructions in the PATCH request have been implemented, the UDR shall respond with 204 No Content response; if some of the modification instructions in the PATCH request have been discarded, and the NF service consumer has included in the supported-feature query parameter the "PatchReport" feature number, the UDR shall respond with PatchResult.				

5.3 Custom Operations without associated resources

None in this release of the specification.

5.4 Notifications

5.4.1 General

This clause specifies the notifications provided by the Nadm_DM service.

The delivery of notifications shall be supported as specified in clause 6.2 of TS 29.500 [4] for Server-initiated communication.

Table 5.4.1-1: Notifications overview

Notification	Callback URI	HTTP method or custom operation	Description (service operation)
AF Authorization Data Change Notification	{notifyCallbackUri} (NF Service Consumer provided callback reference)	POST	Notify changes of AF Authorization Data

5.4.2 Notify changes of AF Authorization Data

5.4.2.1 Description

If the NF Service Consumer (e.g AIOTF) has provided the callback URI for getting notified about changes of AF Authorization Data, the ADM shall notify the NF Service Consumer when the AF Authorization Data is changed.

5.4.2.2 Notification Definition

The POST method shall be used for AF Authorization Data Change Notification, and the URI shall be the callback reference provided by the NF Service Consumer during the subscription to this notification.

Callback URI: **{notifyCallbackUri}**

Support of URI query parameters is specified in table 5.4.2.2-1.

Table 5.4.2.2-1: URI query parameters supported by the POST method

Name	Data type	P	Cardinality	Description
n/a				

Support of request data structures is specified in table 5.4.2.2-2, and support of response data structures and response codes is specified in table 5.4.2-3.

Table 5.4.2.2-2: Data structures supported by the POST Request Body

Data type	P	Cardinality	Description
AfAuthorizationDataChangeNotification	M	1	Representation of AF Authorization Data Change Notification

Table 5.4.2.2-3: Data structures supported by the POST Response Body

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Successful notification of the AF Authorization Data Change Notification
NOTE: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] other than those specified in the table above also apply, with a ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).				

5.5 Data Model

5.5.1 General

This clause specifies the application data model supported by the API.

Table 5.5.1-1 specifies the data types defined for the Nudr_DataRepository for AIoT data service-based interface protocol.

Table 5.5.1-1: Nudr_DataRepository specific Data Types for AIoT data

Data type	Section defined	Description	Applicability
TidInformation	5.5.2.4	T-ID Information	

Table 5.5.1-2 specifies data types re-used by the Nudr_DataRepository for AIoT Data service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nudr_DataRepository service based interface for Ambient IoT data.

Table 5.5.1-2: Nudr_DataRepository re-used Data Types for AIoT data

Data type	Reference	Comments	Applicability
AiotDevPermId	TS 29.571 [18]	AIoT device permanent identifier	
AiotDevProfileData	TS 29.369 [19]	AIoT Device Profile Data	
AiotDevProfileDataList	TS 29.369 [19]	AIoT Device Profile Data of bundled AIoT Devices	
PatchItem	TS 29.571 [18]	Data structure used for JSON patch.	
PatchResult	TS 29.571 [18]	Patch Result	
ProblemDetails	TS 29.571 [18]	Used in error responses to provide more detailed information about an error.	
SupportedFeatures	TS 29.571 [18]	see TS 29.500 [4] clause 6.6	
AfAuthorizationData	TS 29.369 [19]	AF Authorization Data	
AfId	TS 29.369 [19]	AF ID	
AfAuthorizationDataSubscription	TS 29.369 [19]	AF Authorization Data Subscription	
AfAuthorizationDataChangeNotification	TS 29.369 [19]	AF Authorization Data Change Notification	
TidHandlingInformation	TS 29.369 [19]	T-ID Handling Information	
Tid	TS 29.369 [19]	AIoT Device Temporary Identifier (T-ID)	

5.5.2 Structured data types

5.5.2.1 Introduction

This clause defines the structures to be used in resource representations.

5.5.2.2 Void

5.5.2.3 Void

5.5.2.4 Type: TidInformation

Table 5.5.2.4-1: Definition of type TidInformation

Attribute name	Data type	P	Cardinality	Description	Applicability
tid	Tid	O	0..1	Current Temporary Identifier (T-ID_n) as specified in TS 33.369 [18].	
tidHandlingInfo	TidHandlingInformation	O	0..1	T-ID handling information as specified in clause 5.4.3 of TS 33.369 [18].	

5.5.3 Simple data types and enumerations

5.5.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

5.6 Error handling

Table 5.6-1 lists common response body data structures used within the nudr-dr (Nudr_DataRepository) API.

Table 5.6-1: Common Response Body Data Structures

Data type	P	Cardinality	Response codes	Description
ProblemDetails	O	0..1	4xx, 5xx responses	For unsuccessful status codes, the UDR may provide detailed information.
NOTE: In addition common data structures as defined in TS 29.500 [4] are supported.				

The application error handling shall follow TS 29.504 [15] clause 6.1.6 unless explicitly specified in this document.

5.7 Feature negotiation

The optional features in table 6.1.8-1 of TS 29.504 [15] are defined for the Nudr_DataRepository API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of TS 29.500 [4].

Annex A (normative): OpenAPI specification

A.1 General

This Annex specifies the formal definition of the API(s) defined in the present specification. It consists of OpenAPI specifications in YAML format.

This Annex takes precedence when being discrepant to other parts of the specification with respect to the encoding of information elements and methods within the API(s).

NOTE 1: The semantics and procedures, as well as conditions, e.g. for the applicability and allowed combinations of attributes or values, not expressed in the OpenAPI definitions but defined in other parts of the specification also apply.

Informative copies of the OpenAPI specification files contained in this 3GPP Technical Specification are available on a Git-based repository that uses the GitLab software version control system (see clause 5.3.1 of TS 29.501 [5] and clause 5B of TR 21.900 [7]).

A.2 Nudr_DataRepository API for AIoT data API

For the purpose of referencing entities in the Open API file defined in this Annex, it shall be assumed that this Open API file is contained in a physical file named "TS29506_Aiot_Device_Profile_Data.yaml".

```
openapi: 3.0.0

info:
  version: '-'
  title: Unified Data Repository Service API file for Ambient IoT data
  description: |
    The API version is defined in 3GPP TS 29.504
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: >
    3GPP TS 29.506 V19.3.0; 5G System; Usage of the Unified Data Repository Service for AIoT Data
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.506/'

paths:
  /aiot-data/aiot-device-profile-data/{aiotDevPermId}:
    parameters:
     - name: aiotDevPermId
       in: path
       required: true
       schema:
         $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotDevPermId'
    get:
      summary: Retrieves the AIoT device profile data for a given AIoT device permanent identifier
      operationId: ReadAiotDeviceProfileData
      tags:
        - AiotDeviceProfileData (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:aiot-data:aiot-device-profile-data
          - nudr-dr:aiot-data:aiot-device-profile-data:aiotDevPermId:read
      responses:
        '200':
          description: >
            Upon success, a response body containing AIoT device profile data shall be returned.
          content:
            application/json:
              schema:
                $ref: 'TS29369_Nadm_DM.yaml#/components/schemas/AiotDevProfileData'
```

```
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        description: Not Found
        content:
          application/problem+json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  patch:
    summary: Modify the AIoT device profile data for a given AIoT device permanent identifier.
    operationId: UpdateAiotDeviceProfileData
    tags:
      - AiotDeviceProfileData (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:aiot-data:aiot-device-profile-data
        - nudr-dr:aiot-data:aiot-device-profile-data:aiotDevPermId:modify
    parameters:
      - name: supported-features
        in: query
        description: Supported Features
        required: false
        schema:
           $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    requestBody:
      required: true
      content:
        application/json-patch+json:
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchItem'
    responses:
      '200':
        description: Expected response to a valid request.
        content:
          application/json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchResult'
      '204':
        description: No content. Response to successful modification.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        description: Not Found
        content:
          application/problem+json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
```

```
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /aiot-data/aiot-device-profile-data:
    get:
      summary: Retrieves the AIoT device profile data for bundled AIoT devices
      operationId: ReadBundledAiotDeviceProfileData
      tags:
        - BundledAiotDeviceProfileData (Document)
      parameters:
        - name: requester-aiot-devices-id
          in: query
          description: a list of AIoT Device Permanent ID
          required: true
          style: form
          explode: false
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotDevPermId'
            minItems: 1
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:aiot-data:aiot-device-profile-data
          - nudr-dr:aiot-data:aiot-device-profile-data:read
      responses:
        '200':
          description: >
            Upon success, a response body containing bundled AIoT device profile data shall be
returned.
          content:
            application/json:
              schema:
                $ref: 'TS29369_Nadm_DM.yaml#/components/schemas/AiotDevProfileDataList'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

    patch:
      summary: Modify the AIoT device profile data for bundled AIoT devices.
      operationId: UpdateBundledAiotDeviceProfileData
      tags:
        - BundledAiotDeviceProfileData (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:aiot-data:aiot-device-profile-data
          - nudr-dr:aiot-data:aiot-device-profile-data:modify
      parameters:
        - name: supported-features
          in: query
          description: Supported Features
          required: true
```

```
        schema:
           $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    requestBody:
      required: true
      content:
        application/merge-patch+json:
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchItem'
    responses:
      '200':
        description: Expected response to a valid request.
        content:
          application/json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchResult'
      '204':
        description: No content. Response to successful modification.
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'


  /aiot-data/af-authorization-data:
    get:
      summary: get AF Authorization Data
      operationId: Get AF Authorization Data
      tags:
        - AF Authorization Data Retrieval
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr:aiot-data:af-authorization-data
          - nudr-dr:aiot-data:af-authorization-data:read
      parameters:
        - name: af-id
          in: query
          description: AF ID
          required: false
          schema:
            $ref: 'TS29369_Nadm_DM.yaml#/components/schemas/AfId'
      responses:
        '200':
          description: Expected response to a valid request
          content:
            application/json:
              schema:
                $ref: 'TS29369_Nadm_DM.yaml#/components/schemas/AfAuthorizationData'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
```

```
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error

  /aiot-data/af-authorization-data/subscriptions:
    post:
      summary: Create a subscription of AF Authorization Data Changes
      operationId: Subscribe AF Authorization Data Change
      tags:
        - Create a subscription of AF Authorization Data Changes
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:aiot-data:af-authorization-data:subscriptions:create
      requestBody:
        content:
          application/json:
            schema:
              $ref: 'TS29369_Nadm_DM.yaml#/components/schemas/AfAuthorizationDataSubscription'
        required: true

      callbacks:
        afAuthorizationDataChangeNotification:
          '{$request.body#/notifyCalblackUri}':
            post:
              requestBody:  # contents of the callback message
                required: true
                content:
                  application/json:
                    schema:
                      $ref:
'TS29369_Nadm_DM.yaml#/components/schemas/AfAuthorizationDataChangeNotification'
              responses:
                '204':
                  description: successful notification
                '400':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/400'
                '403':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/403'
                '404':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/404'
                '411':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/411'
                '413':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/413'
                '415':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/415'
                '429':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/429'
                '500':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/500'
                '502':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/502'
                '503':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      responses:
        '201':
          description: successful creation of the subscription
          content:
            application/json:
              schema:
                $ref: 'TS29369_Nadm_DM.yaml#/components/schemas/AfAuthorizationDataSubscription'
          headers:
            Location:
              description: >
                'Contains the URI of the newly created subscription, according to the structure:
```

```
              {apiRoot}/nudr-dr/<apiVersion>/aiot-data/af-authorization-
data/subscriptions/{subId}'
            required: true
            schema:
              type: string
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      '504':
        $ref: 'TS29571_CommonData.yaml#/components/responses/504'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /aiot-data/af-authorization-data/subscriptions/{subId}:
    patch:
      summary: Modify a subscription of AF Authorization Data Changes
      operationId: Modify Subscription of AF Authorization Data Change
      tags:
        - Modify a subscription of AF Authorization Data Changes
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:aiot-data:af-authorization-data:subscriptions:modify
      requestBody:
        content:
          application/json-patch+json:
            schema:
              type: array
              items:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchItem'
        required: true
      responses:
        '200':
          description: Expected response to partially successful patch
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchResult'
        '204':
          description: Expected response to successful patch
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
```

```
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  delete:
    summary: Delete a subscription of AF Authorization Data Changes
    operationId: Unsubscribe to AF Authorization Data Change
    tags:
      - Delete a subscription of AF Authorization Data Changes
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:aiot-data:af-authorization-data:subscriptions:modify
    responses:
      '204':
        description: Expected response to successful deletion
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '501':
        $ref: 'TS29571_CommonData.yaml#/components/responses/501'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'


/aiot-data/tid-info-store/{aiotDevPermId}:
  parameters:
   - name: aiotDevPermId
     in: path
     required: true
     schema:
       $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotDevPermId'
  get:
    summary: Retrieves the stored T-ID information for a given AIoT device
    operationId: ReadTidInformation
    tags:
      - IndividualTidInformation (Document)
    security:
      - {}
      - oAuth2ClientCredentials:
        - nudr-dr
        - nudr-dr:aiot-data:tid-info-store
        - nudr-dr:aiot-data:tid-info-store:aiotDevPermId:read
    responses:
      '200':
        description: >
          Upon success, a response body containing the stored T-ID information shall be returned.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/TidInformation'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
```

```
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    patch:
      summary: Update the stored T-ID information for a given AIoT device.
      operationId: UpdateTidInformation
      tags:
        - IndividualTidInformation (Document)
      security:
        - {}
        - oAuth2ClientCredentials:
          - nudr-dr
          - nudr-dr:aiot-data:tid-info-store
          - nudr-dr:aiot-data:tid-info-store:aiotDevPermId:modify
      parameters:
        - name: supported-features
          in: query
          description: Supported Features
          required: false
          schema:
             $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      requestBody:
        required: true
        content:
          application/json-patch+json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchItem'
      responses:
        '200':
          description: Expected response to a valid request.
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchResult'
        '204':
          description: No content. Response to successful modification.
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:

  schemas:

# STRUCTURED TYPES
    TidInformation:
```

```
      description: Contains the stored T-ID information for an AIoT device.
      type: object
      properties:
        tid:
          $ref: 'TS29369_Nadm_Sec.yaml#/components/schemas/Tid'
        tidHandlingInformation:
          $ref: 'TS29369_Nadm_Sec.yaml#/components/schemas/TidHandlingInformation'


# ENUMS:
```

Annex B (informative): Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2025-05	CT4#129	C4-252334				TS skeleton	0.0.0
2025-08	CT4#130	C4-253541				Inclusion of pCRs agreed at CT4#130, including C4-253414, C4-253528, C4-253562, C4-253574.	0.1.0
2025-09	CT#109	CP-252183				Presented for information and approval	1.0.0
2025-09	CT#109					Approved in TSG CT#109	19.0.0
2025-12	CT#110	CP-253160	0001		F	Correct the description of lastKnownAiotfInfo	19.1.0
2025-12	CT#110	CP-253160	0003	1	F	Corrections on UDR service for AIoT	19.1.0
2025-12	CT#110	CP-253167	0004		F	API version and External doc update	19.1.0
2026-03	CT#111	CP-260036	0006		F	Correction on the security part	19.2.0
2026-03	CT#111	CP-260036	0007		F	Update on AIoT Patch	19.2.0
2026-03	CT#111	CP-260036	0005	2	F	Update AiotDeviceProfileData to align with 29369	19.2.0
2026-03	CT#111	CP-260036	0009	2	F	Enable bundled AIoT Device Profile Data retrieval and update	19.2.0
2026-03	CT#111	CP-260040	0010		F	API version and External doc update	19.2.0
2026-06	CT#112	CP-261054	0011	3	F	Nudr Support for Nadm DM Subscription Operations	19.3.0
2026-06	CT#112	CP-261054	0012	2	F	Separate T-ID from Device Profile Data	19.3.0
2026-06	CT#112	CP-261088	0013		F	API version and External doc update	19.3.0

History

Version	Date	Status
V19.0.0	January 2026	Publication
V19.1.0	February 2026	Publication
V19.2.0	March 2026	Publication
V19.3.0	August 2026	Publication